Supplementary Material for “Information-Set Conditional Integrity Auditing for Hybrid Quantum-Classical Kernel Workflows”

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1 Purpose and Claim Boundary

This supplement records the complete experimental design, coverage contract, validation checks, and reproduction mapping behind the main article. It adds no QPU, calibrated-noise, scheduling, provider-security, multi-tenancy, or operational Fleet Management evidence. The exact claims remain bounded to data-to-evaluation interventions, ideal-statevector quantum kernels, controlled binomial shot emulation, and a local executable contract.

2 Frozen Experimental Design

2.1 Expansion Environments

The expansion comprises eight fixed environments (Table 1). “OOD” means a prespecified source-to-target construction, not a sampled deployment population. Every environment uses five split seeds and two nested model seeds. All evaluate projected dimensions 8, 10, and 12. The CICIDS scale gate uses 256/256 train/evaluation samples; all others use 128/128. The two additional ID datasets retain all empirically distinct quantum profiles. Scale and temporal gates compare SVC-RBF with QSVC-ZZ.

Each staged table contains 3,000 balanced binary rows before experimental subsampling. Numeric non-finite values are replaced during staging; categorical fields are excluded. These choices produce a controlled integrity audit but do not reproduce operational prevalence. Several temporal targets yield clean performance near chance, limiting the impact that later interventions can express; the exact information-set invariances do not depend on clean headroom.

2.2 Perturbation Suite

The paper-core suite crosses six mechanisms with nominal strengths 0.02, 0.05, and 0.10 (Table 2). Artifact identifiers retain the historical family string `target_shift`; the scientific text uses *evaluation-label intervention* because these rows do not instantiate the statistical label-shift problem.

The feature mechanisms may alter representation, scores, predictions, and the reported conclusion. Both label mechanisms keep the feature representation, scores, and predictions fixed. Prior-preserving flips also retain the empirical class histogram. Equal mechanism weighting is a deliberately transparent stress profile, not an estimate of real incident frequency.

3 Information-Set Coverage Contract

Table 3 records what each evidence regime can establish in the implemented study. Structural blindness means that the sensor’s complete input is unchanged by construction; it is stronger and narrower than observing zero empirical response in a finite sample.

Every machine-readable coverage record contains seven fields:

1) workflow boundary;
2) protected asset;
3) adversary or failure capability, including fixed elements;
4) available information regime;
5) sensor, reference, threshold, and false-alarm budget;
6) covered perturbations and residual blind region; and
7) response or countermeasure.

This record prevents a sensor response from being promoted into a global security claim.

4 Deduplication and Statistical Units

Gate 1 emits 7,980 raw model-intervention rows. Different quantum-map jobs repeat the same classical SVC pipeline, producing 3,420 repeated rows. The builder compares every repeated key across performance, impact, and raw signal columns before retaining one representative; 4,560 unique observations remain.

The expansion emits 13,680 raw rows from 360 completed CSV/JSON job pairs. Its multi-map ID gates contain 2,280 repeated SVC rows, again accepted as duplicates only after numerical identity checks; 11,400 unique observations remain. The strict builder fails if a job is missing, an environment contains an unexpected model/dimension/seed/attack design, or a duplicated row disagrees.

Coverage counts use prespecified cells as their denominators. For the secondary suite-average model profile, nested model-seed values are averaged inside each of five split seeds. The split seed is the inferential unit, and the interval is

$$\bar{d} \pm t_{0.975,4}\frac{s_d}{\sqrt{5}}, \tag{1}$$

where d is the within-split ZZ-minus-SVC impact difference. The intervals are not multiplicity-adjusted and do not

TABLE 1
Prespecified expansion environments.

ID	Environment	Training source	Evaluation source	Samples	Profiles
E1	CICIDS scale/ID	Frozen CICIDS subset	Same staged source	256/256	SVC, ZZ
E2	UNSW-NB15 ID	Balanced staged UNSW	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ-equivalent
E3	ToN-IoT ID	Balanced staged ToN-IoT	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ-equivalent
E4	CICIDS Tue→Wed	Tuesday traffic	Wednesday traffic	128/128	SVC, ZZ
E5	CICIDS Tue→Fri-PortScan	Tuesday traffic	Friday PortScan traffic	128/128	SVC, ZZ
E6	CICIDS Wed→Thu-Web	Wednesday traffic	Thursday Web traffic	128/128	SVC, ZZ
E7	CICIDS Wed→Fri-AM	Wednesday traffic	Friday morning traffic	128/128	SVC, ZZ
E8	UNSW temporal OOD	Prespecified UNSW source period	Prespecified UNSW target period	128/128	SVC, ZZ

TABLE 2
Frozen 18-condition intervention suite.

IDs	Mechanism	Strengths
1–3	Feature sign flip	0.02, 0.05, 0.10
4–6	Feature-wise mean shift	0.02, 0.05, 0.10
7–9	Scaling drift	0.02, 0.05, 0.10
10–12	Feature dropout	0.02, 0.05, 0.10
13–15	Prior-preserving evaluation-label flip	0.02, 0.05, 0.10
16–18	Random evaluation-label flip	0.02, 0.05, 0.10

support population transport. Results over 20 split-model combinations are sensitivity analyses only.

5 Exact-Statevector Engine Validation

The expansion evaluator replaces pairwise compute-uncompute execution with a mathematically equivalent ideal-statevector calculation. It prepares each unique state $|\phi(x)\rangle$ once and forms

$$K(x,y) = |\langle \phi(x) \mid \phi(y) \rangle|^2 \tag{2}$$

by matrix multiplication. Symmetric blocks use the same eigenvalue-clipping PSD repair as the frozen Qiskit reference.

Validation has four layers:

1) symmetric and cross-kernel unit matrices agree with the reference at relative and absolute tolerance 10^{-10};
2) symmetry, unit diagonal, numerical range, cache bounds, and cache reuse are unit-tested;
3) a CICIDS 128/128 end-to-end replay (split/model seed 42, dimension 8, ZZ, paper-core suite) preserves every categorical, discrete performance, impact, and integrity-signal endpoint; and
4) the prespecified 256/256 pilot completes.

The two end-to-end outputs each have 38 rows and 126 columns. Only ROC-AUC and its derived drops differ, with maximum absolute difference 1.221896×10^{-4}, approximately one half-tie increment for 64 positive and 64 negative evaluation examples. This supports an explicit 2×10^{-4} ROC-AUC tolerance, not bitwise identity of continuous scores. The engine is tagged `exact_statevector`; it supplies no finite-shot, noise-model, or hardware evidence.

6 Quantum Gate: Conditions and Coverage

The quantum gate crosses five split seeds, dimensions 4/6/8, and 11 conditions, giving 165 cells. Table 4 reports response fractions over the 15 split-dimension cells per condition. A value of one means nonzero raw response in all 15 cells at the frozen tolerance; it is not calibrated power.

All nine frozen acceptance checks pass: 165 cells are present; clean evidence is invariant; benign transpilation and common-unitary rewrite alter provenance but preserve semantic kernels and outputs; every data-dependent circuit mutation changes the semantic kernel; symmetry and diagonal sensors catch their respective malformed edits; the PSD-preserving mixture passes algebra but is exposed by trusted provenance/semantics; and both shot estimators produce nonzero repeat discrepancy in every cell.

7 Executable Contract Details

The prototype evaluates four ordered contracts:

1) **Input/preprocessing**: input-array and preprocessing-declaration hashes; mismatch blocks.
2) **Circuit/kernel**: circuit/kernel provenance, semantic probes, symmetry, diagonal, eigenvalue, and PSD-repair evidence; an equivalent rewrite passes only if explicitly approved.
3) **Execution/result**: declared execution metadata, repeated- estimation discrepancy, and prediction provenance; approved stochastic execution is held for adjudication.
4) **Evaluation/report**: item-level labels are bound to predictions and balanced accuracy is recomputed; prior-preserving corruption blocks even when the marginal is identical.

Contracts form an ordered SHA-256 chain. Later modification is detectable, but the chain does not authenticate a provider because it is not digitally signed. The response lattice is deliberately fail-closed: all passes yield `allow`; at least one review and no failure yields `hold`; any failure yields `block`. Six prespecified scenarios and all six chains pass eight strict acceptance checks in the frozen artifact.

TABLE 3
Evidence regimes, implemented sensors, and demonstrated blind regions.

Regime	Evidence	Implemented sensor family	Exact/empirical blind region	Minimum closing evidence in this study
$\mathcal{I}_X$	Features	Feature JSD, MMD, KS rejection	Every label-only change is structurally blind	Trusted label or item provenance
$\mathcal{I}_{XF}$	Features, scores, predictions	$\mathcal{I}_X$ plus score JSD and prediction disagreement	Every label-only change is structurally blind for a fixed predictor	Item-aligned outcome-prediction evidence
$\mathcal{I}_{Y_m}$	Label class counts/marginal	Prior delta and label JSD	Prior-preserving item-level relabeling is structurally blind	Trusted item-level label provenance
$\mathcal{I}_{XFY}$	Item-aligned features, predictions, outcomes	Prediction-rate, disagreement, prediction-distribution, confusion-profile deltas	Closes every positive label-path case executed, conditional on trusted outcomes	Independent authentication of the joint evidence
$\mathcal{I}_Q$	Circuit, parameters, kernel, estimation evidence	Hashes, semantic probes, algebra, repeated estimation, output changes	Sensor-dependent: hashes overreact to equivalence; algebra misses PSD-preserving substitution; outputs miss sub-decision changes	Layered policy-aware composition

TABLE 4
Quantum-gate sensor response fractions.

Condition	Circuit prov.	Kernel prov.	Semantic kernel	Algebraic kernel	Repeated estimate	Output change
Clean	0.00	0.00	0.00	0.00	0.00	0.00
Benign transpilation	1.00	1.00	0.00	0.00	0.00	0.00
Common-unitary rewrite	1.00	1.00	0.00	0.00	0.00	0.00
Parameterized RZ 0.02	1.00	1.00	1.00	0.00	0.00	0.00
Parameterized RZ 0.10	1.00	1.00	1.00	0.00	0.00	0.13
Feature-map reps 1→2	1.00	1.00	1.00	0.00	0.00	0.73
Asymmetric kernel edit	0.00	1.00	1.00	1.00	0.00	0.00
Kernel diagonal erosion	0.00	1.00	1.00	1.00	0.00	0.00
PSD-preserving kernel mix	0.00	1.00	1.00	0.00	0.00	0.07
Shot estimator 256	0.00	1.00	1.00	1.00	1.00	0.47
Shot estimator 1,024	0.00	1.00	1.00	1.00	1.00	0.20

8 Reproduction and Evidence Manifest

The recommended read-only verification is:

```
python -m src.experiments.`
verify_publication_artifact `
--root publication/artifact
```

It verifies the artifact-wide SHA-256 manifest, four embedded evidence manifests, 22 outputs, row counts, and primary-claim counts. In the frozen release it reports 96 files, 3,600 expansion label rows, 2,184 positive expansion impacts, 1,440 Gate-1 label rows, and 1,276 positive Gate-1 impacts.

The complete test command is:

```
python -m pytest tests -q -p no:cacheprovider
```

The frozen environment uses Python 3.10.16, NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, Qiskit 2.3.0, and Qiskit Machine Learning 0.9.0. Qiskit is pinned below 3 because migrating deprecated feature-map constructors would change circuit serialization and therefore requires a new evidence version.

The artifact intentionally omits raw benchmark datasets. It includes staging instructions, source locations, source code, locked dependencies, compact derived evidence, figures, figure-source tables, tests, manifests, and expected outputs. Public DOI insertion remains contingent on confirmed authorship, license, and repository visibility; those administrative fields do not alter the scientific results.

9 Claims-Supported / Claims-Excluded Matrix

Supported by the artifact. Auditability is conditional on evidence exposed at a declared boundary; model-output and evaluation-conclusion impacts can have different causal loci; feature/prediction sensors are exactly blind to label-only changes, whereas label marginals are blind to prior-preserving relabeling; complementary quantum-workflow sensors close different simulator-gate blind regions; and the local prototype executes policy-aware, fail-closed `allow/hold/block` composition.

Explicitly excluded. The evidence does not estimate attack prevalence or population-level deployment incidence, establish universal superiority or vulnerability of quantum kernels, or causally isolate feature-map geometry from preprocessing and clean headroom. It provides no evidence about QPU security, calibrated device noise, scheduling, provider integrity, multi-tenancy, side channels, production readiness, provider authentication, non-repudiation, persistence, incident handling, or operational Fleet Management.